

Solution Requirements

Date	26-06-2026
Team ID	SWTID-2026-3845
Project Name	SMART LENDER
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	The system shall provide a web form for applicants to enter personal and financial details (gender, marital, credit history, property area).	High
2	Data Validation & Preprocessing	The system shall validate input fields and preprocess data into the format required by the model.	High
3	Model Training & Selection	<i>The system shall train and evaluate classification algorithms and select the best-performing model.</i>	High

4	Model Persistence & Loading	The system shall save the selected best-performing model (XGBoost) and load it for real-time predictions.	High
5	Real-Time Prediction	The system shall generate a loan approval/rejection prediction with a confidence score immediately after an applicant submits their details.	High
6	Risk Classification	The system shall classify applicants as low-risk or high-risk based on the model's prediction to support review decisions.	Medium
7	Batch Evaluation & Reporting	The system shall allow credit officers/analysts to evaluate multiple applicants at once and generate prediction reports/summaries.	Medium
8	Record Storage & Audit Logging	The system shall store prediction outcomes in a database, maintaining logs for compliance and audit purposes.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance	The system shall return a loan prediction result within an acceptable	Prediction result displayed within 3 seconds
2	Accuracy	The deployed model shall maintain a high level of predictive accuracy on unseen applicant data	≥ 80% testing accuracy (XGBoost: 81.1%)
3	Usability	The web interface shall be simple and intuitive for both applicants and bank staff with minimal training.	New users complete a prediction request without assistance in under 2 minutes
4	Scalability	The system shall support an increasing number of concurrent users and batch evaluation	Handles at least 100 concurrent users on Render
5	Availability	The system shall be accessible with minimal downtime for continuous loan processing.	≥ 99% uptime
6	Security	The system shall protect sensitive applicant data (income, credit history) from unauthorized access.	All data transmitted over HTTPS; role-based access for staff features